

KYAMBOGO UNIVERSITY
COURSE WORK.

**FACULTY: SCHOOL OF COMPUTIN AND
INFORMATION SCIENCE**

**COURSE UNIT. INTRODUCTION TO
PROGRAMMING**

COURSE CODE: SIS1201

YEAR: ONE SEMESTER: TWO

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NO.1

Computer hardware refers to the physical and tangible part of a computer which helps in input and output of data and include keyboard, scanner, monitors, servers, routers.

While

A computer software are programs and applications that run on a computer hardware.

A computer hardware and software work together to run a program in a way that the software gives instructions to the hardware basically the cpu on what task to do and then the hardware performs them according to the given instruction.

When a program is opened, it is loaded into memory and processed by the cpu and the output is then shown on the screen.

NO.2

Developing a C program involves three main stages.

The first stage is writing the program. This is where the programmer writes the code using a text editor following C language rules.

The second stage is compiling. In this stage, the program is checked for errors using a compiler. If there are mistakes, they are corrected before moving on.

The third stage is running the program. After fixing errors, I run the program to see the output and confirm if it works the way I expected.

NO.3

A selection structure is when a program makes a decision depending on a condition. It allows a program to choose one action instead of another. For example, it can decide what to do if something is true or false.

In my daily life, I make decision using selection structures without noticing. For example, when I wake up, I decide what to wear depending on the weather. If it is raining, I carry a jacket, but if it is sunny, I dress lightly.

If weather is raining

Wear jacket else wear light clothes

```
1  /*this program shows two integers and their outputs*/
2  #include <stdio.h>
3  int main(){
4      int numA, numB;
5      int sum, difference, product;
6      printf("Enter the first number: ");
7      scanf("%d",&numA);
8      printf("Enter second number: ");
9      scanf("%d",&numB);
10     sum =numA+ numB;
11     difference =numA-numB;
12     product =numA* numB;
13     printf("sum=%d\n", sum);
14     printf("difference=%d\n",difference);
15     printf("product=%d\n", product);
16     return 0;

```

Compile Log Debug Find Results Close

```
Enter the first number: 12
Enter second number: 5
sum=17
difference=7
product=60

-----
Process exited after 13.25 seconds with return value 0
Press any key to continue . . .

```

```
1 // FILE: 1001_1001.c
2 int main(){
3     int firstnum, secondnum;
4     printf("enter the first integer; ");
5     scanf("%d", &firstnum);
6     printf("enter the second integer: ");
7     scanf("%d", &secondnum);
8     if (firstnum > secondnum){
9         printf("%d is greater than %d\n", firstnum, secondnum);
10    }else if (firstnum < secondnum){
11        printf("%d is less than %d\n", firstnum, secondnum);
12    }else{
13        printf("both numbers are equal: %d=%d\n", firstnum, secondnum);
14    }
15    return 0;
16 }
```

Compile Log Debug Find Results Close

```
enter the first integer; 7
enter the second integer: 3
7 is greater than 3

-----
Process exited after 21.56 seconds with return value 0
Press any key to continue . . .
```

```

1  #include <stdio.h>
2  int main(){
3      int score;
4      printf("Enter the students exam score (0-100): ");
5      scanf("%d",&score);
6
7      if (score >= 80){
8          printf("Distinction.\n");
9          if (score == 100)
10             {printf("Full scholarship awarded!\n ");}
11         else
12             {printf("Partial sholarship awarded!\n ");}
13     }
14     else if(score >= 50 && score <= 79){
15         printf("Passed.\n");
16         if (score < 60){
17             printf("Note: You are on the caution list.\n "); }
18     }
19
20     else {
21         if(score < 50)
22             { printf("Failed\n"); }
23         if(score < 20)
24             { printf("Required to repeat the semester\n"); }
25         else{ printf("Eligible for retake exam.\n"); }
26     }

```

```

Enter the students exam score (0-100): 05
Failed
Required to repeat the semester

-----
Process exited after 19.77 seconds with return value 0
Press any key to continue . . .

```